

# Sara Klaasen

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📍 Canberra, Australia

🌐 [saraklaasen.github.io](https://saraklaasen.github.io)

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## Education

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|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| <b>PhD</b> | <b>ETH Zurich</b> , Geophysics <ul style="list-style-type: none"><li>Eidgenössische Technische Hochschule Zürich (ETH Zürich, Swiss Federal Institute of Technology Zurich)</li><li>ETH medal</li></ul> | Zurich, Switzerland<br>2020 – 2025     |
| <b>Msc</b> | <b>ETH Zurich</b> , Geophysics                                                                                                                                                                          | Zurich, Switzerland<br>2018 – 2020     |
| <b>Bsc</b> | <b>University College Roosevelt (UCR)</b> , Liberal Arts and Sciences <ul style="list-style-type: none"><li>Magna cum Laude</li><li>Major in Earth sciences, biology, mathematics and physics</li></ul> | Middelburg, Netherlands<br>2014 – 2017 |

## Experience

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| <b>Australian National University</b> , Postdoctoral researcher<br>Researching DAS projects in New Zealand. <ul style="list-style-type: none"><li>Imaging Taupo Volcano.</li></ul>                       | Canberra, Australia<br>2026 – present<br>1 year   |
| <b>ETH Zurich</b> , Postdoctoral researcher                                                                                                                                                              | Zurich, Switzerland<br>2025 – 2026<br>1 year      |
| <b>Sport, Recreatie en Onderwijsvoorzieningen (SRO)</b> , Admin<br>Online administration                                                                                                                 | Haarlem, Netherlands<br>2017 – 2017<br>1 year     |
| <b>Provinciale Zeeuwse Courant (PZC)</b> , Online editor <ul style="list-style-type: none"><li>Online content management of the provincial newspaper</li><li>Writing and editing news articles</li></ul> | Vlissingen, Netherlands<br>2015 – 2017<br>2 years |

## Awards

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| <b>ETH Medal</b><br>The ETH Medal is given to outstanding doctoral theses (2k CHF).<br>ETH Zurich          | 2026 |
| <b>Outstanding Student Presentation Award (OSPA)</b><br>Seismological Society of America Photonics Meeting | 2024 |

## Grants

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| <b>Swiss National Science Foundation (SNF) postdoc.mobility grant</b><br>Grant to fund a 2-year postdoc position (136k CHF)<br>Swiss National Science Foundation | 2025 |
| <b>Women-in-STEM grant</b><br>Grant for peer-mentoring group (2k CHF).<br>Fix the Leaky Pipeline program, ETH Zürich                                             | 2021 |

## **VSBfonds grant**

2018

Grant to support a Masters programme abroad (10k EUR).  
VSBfonds

## **Research collaborations**

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### **Australian Research Council (ARC) Centre of Excellence in Optical Microcombs for Break-through Science (COMBS)**

2026 – present

Associate investigator

[www.combs.org.au](http://www.combs.org.au)

### **Seismicity on Venus: Prediction and Detection**

2022 – 2024

International Space Science Institute (ISSI) international team.

ECR member

[teams.issibern.ch/seismicityonvenus](https://teams.issibern.ch/seismicityonvenus)

## **Teaching experience**

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### **Data Analysis and Visualisation with Python in Earth Sciences**

2022 – 2025

Bachelor course at ETH Zürich

Teaching assistant

### **Geophysical field lab**

2021 – 2023

Bachelor fieldcourse at ETH Zürich

Teaching assistant

### **Dynamische Erde I**

2020 – 2023

Annual ten minute lectures about current geophysical applications during the Bachelor course at ETH Zürich.

Guest lecturer

## **Research Output**

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### **Book Chapters**

#### **The leading edge of volcanology: Advances and innovations in monitoring and interpreting volcanic activity**

2027

Book chapter submitted to "The encyclopedia of Volcanoes, Third Edition". Chapter has been accepted.

Poland, Michael P and de Zeeuw-van Dalfsen, Elske and Aizawa, Koki and Klaasen, Sara and D'Arcy, Fiona and Nicholson, Emma J and Anderson, Kye  
(Elsevier)

#### **Fiber optic sensing for volcano monitoring and imaging volcanic processes**

2025

Book chapter in Modern Volcano Monitoring

Philippe Jousset, Gilda Currenti, Shane Murphy, Eva Eibl, Corentin Caudron, Lise Retailleau, Christopher Wollin, Sara Klaasen, Takeshi Nishimura, Andreas Fichtner, Zack Spica, Arnaud Lemarchand, Charlotte Krawczyk

[doi.org/10.1007/978-3-031-86841-2\\_17](https://doi.org/10.1007/978-3-031-86841-2_17) (Springer)

### **Journal Articles**

#### **Submarine seismicity monitoring with distributed acoustic sensing near Santorini and Kolumbo Volcano**

2026

Sara Klaasen, Thomas Hudson, Sanchit Sachdeva, Paraskevi Nomikou, Andreas Fichtner

[doi.org/10.26443/seismica.v5i1.1845](https://doi.org/10.26443/seismica.v5i1.1845) (Seismica)

- Seismic constraints on glacier density** 2025  
Ariane Lanteri, Scott Keating, Lars Gebraad, Sara Klaasen, Marta Pienkowska-Cote, Olaf Eisen, Andrea Zunino, Kristín Jónsdóttir, Coen Hofstede, Dimitri Zigone, Andreas Fichtner  
[doi.org/10.1038/s41598-025-26440-z](https://doi.org/10.1038/s41598-025-26440-z) (Scientific Reports)
- The Seismogenic Thickness of Venus** 2025  
Julia Maia, Ana-Catalina Plesa, Iris van Zelst, Richard Ghail, Anna Gülcher, Mark Panning, Sven Peter Näsholm, Barbara de Toffoli, Anna Horleston, Krystyna Smolinski, Sara Klaasen, Robert Herrick, Raphael Garcia  
[doi.org/10.1038/s41598-025-26440-z](https://doi.org/10.1038/s41598-025-26440-z) (Journal of Geophysical Research: Planets)
- Towards a widely applicable earthquake detection algorithm for fibreoptic and hybrid fibreoptic-seismometer networks** 2025  
Thomas Hudson, Sara Klaasen, Olivier Fontaine, Conor Bacon, Kristin Jonsdottir, Andreas Fichtner  
[doi.org/10.1093/gji/ggae459](https://doi.org/10.1093/gji/ggae459) (Geophysical Journal International)
- An illustrated guide to: Distributed and integrated fibre-optic sensing in seismology** 2024  
Andreas Fichtner, Fabian Walter, Patrick Paitz, Sara Klaasen, Daniel Bowden, Sebastian Noe, Nils Müller, Dominik Husmann, Jacques Morel  
[dx.doi.org/10.1016/j.eqsc.2024.09.006](https://dx.doi.org/10.1016/j.eqsc.2024.09.006) (Earthquake Sciences)
- Seismic wave detectability on Venus using ground deformation sensors, infrasound sensors on balloons and airglow imagers** 2024  
Raphael Garcia, Iris van Zelst, Taichi Kawamura, Sven Peter Näsholm, Anna Horleston, Sara Klaasen, Maxence Lefevre, Celine Marie Solberg, Krystyna Smolinski, Ana-Catalina Plesa, Quentin Brissaud, Julia Maia, Simon Stähler, Philippe Lognonné, Mark Panning, Anna Gülcher, Richard Ghail, Barbara de Toffoli  
[doi.org/10.1029/2024EA003670](https://doi.org/10.1029/2024EA003670) (Earth and Space Science)
- Modelling uncertainty in P-wave arrival-times retrieved from DAS data: case-studies from 15 fibre optic cables** 2024  
Emanuele Bozzi, Nicola Piana Agostinetti, Andreas Fichtner, Sara Klaasen, Arantza Ugalde, Biondo Biondi, S Yuan, Torsten Dahm, Marius Isken, Patrick Paitz, Fabian Walter, Alain Baird, Carlos Becerril, Takeshi Nishimura, Junzhu Shen, Tieyuan Zhu, Gilberto Saccorotti  
[doi.org/10.1093/gji/ggae364](https://doi.org/10.1093/gji/ggae364) (Geophysical Journal International)
- Complex spatial distribution of onset amplitude and waveform correlation: case studies from different DAS experiments** 2024  
Emanuele Bozzi, Gilberto Saccorotti, Nicola Piana Agostinetti, Carlos Becerril, Andreas Fichtner, Sara Klaasen, Takeshi Nishimura, Junzhu Shen, Fabian Walter, Tieyuan Zhu  
[doi.org/10.4430/bgo00458](https://doi.org/10.4430/bgo00458) (Bulletin of geophysics and oceanography)
- Challenges in submarine fiber-optic earthquake monitoring** 2024  
Jonas Igel, Sara Klaasen, Sebastian Noe, Paraskevi Nomikou, Konstantinos Korantzalos, Andreas Fichtner  
[doi.org/10.1029/2024JB029556](https://doi.org/10.1029/2024JB029556) (Journal of Geophysical Research: Solid Earth)
- Subglacial volcano monitoring with fibre-optic sensing: Grímsvötn, Iceland** 2023  
Sara Klaasen, Solvi Thrastarson, Yesim Cubuk-Sabuncu, Kristín Jónsdóttir, Lars Gebraad, Patrick Paitz, Andreas Fichtner  
[doi.org/10.30909/vol.06.02.301311](https://doi.org/10.30909/vol.06.02.301311) (Volcanica)

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| <b>Fiber-optic observation of volcanic tremor through floating ice sheet resonance</b><br>Andreas Fichtner, Sara Klaasen, Solvi Thrastarson, Yesim Cubuk-Sabuncu, Patrick Paitz, Kristin Jonsdottir<br><a href="https://doi.org/10.1785/0320220010">doi.org/10.1785/0320220010</a> (The Seismic Record)                                  | 2022 |
| <b>Distributed acoustic sensing in volcano-glacial environments — Mount Meager, British Columbia</b><br>Sara Klaasen, Patrick Paitz, Nadja Lindner, Jan Dettmer, Andreas Fichtner<br><a href="https://doi.org/10.1029/2021JB022358">doi.org/10.1029/2021JB022358</a> (Journal of Geophysical Research: Solid Earth)                      | 2021 |
| <b>Detecting Seismic Events with Computer Vision: Applications for Fiber-Optic Sensing</b><br>Solvi Thrastarson, Robert Torfason, Sara Klaasen, Patrick Paitz, Yesim Cubuk-Sabuncu, Kristin Jonsdottir, Andreas Fichtner<br><a href="https://doi.org/10.1002/essoar.10509693.1">doi.org/10.1002/essoar.10509693.1</a> (ESS Open Archive) | 2021 |
| <b>First-Author Conference Proceedings</b>                                                                                                                                                                                                                                                                                               |      |
| <b>Modelling months of deformation during the Sundhnúkur eruptions on the Reykjanes Peninsula, Iceland</b><br>Sara Klaasen, Daniel Bowden, Kristin Jonsdottir, Yesim Cubuk-Sabuncu, Jack Maxwell, Meghan Miller, Andreas Fichtner<br>(SSA optical seismology and the next era of seismic sensing)                                        | 2026 |
| <b>Submarine volcano monitoring with Distributed Acoustic Sensing at Kolumbo, Greece</b><br>Sara Klaasen, Thomas Hudson, Paraskevi Nomikou, Andreas Fichtner<br>(IAVCEI General Meeting)                                                                                                                                                 | 2025 |
| <b>Full-Waveform Inversion of DAS data on a subglacial volcano: Grímsvötn, Iceland</b><br>Sara Klaasen, Sebastian Noe, Solvi Thrastarson, Yesim Cubuk-Sabuncu, Kristin Jonsdottir, Andreas Fichtner<br>(IAVCEI General Meeting)                                                                                                          | 2025 |
| <b>Submarine volcano monitoring with Distributed Acoustic Sensing at Kolumbo, Greece</b><br>Sara Klaasen, Thomas Hudson, Paraskevi Nomikou, Andreas Fichtner<br>(SSA General Meeting)                                                                                                                                                    | 2025 |
| <b>Moment Tensor inversion with Full-Waveform Inversion and Distributed Acoustic Sensing on a subglacial volcano: Grímsvötn, Iceland</b><br>Sara Klaasen, Sebastian Noe, Solvi Thrastarson, Yesim Cubuk-Sabuncu, Kristin Jonsdottir, Lars Gebraad, Andreas Fichtner<br>(EGU Copernicus Meetings)                                         | 2024 |
| <b>Submarine volcano monitoring with fiber-optic cables at Santorini</b><br>Sara Klaasen, Paraskevi Nomikou, Konstantinos Karantzas, Jonas Igel, Andreas Fichtner<br>(AGU Annual Meeting)                                                                                                                                                | 2023 |
| <b>Going towards Full Waveform Inversion (FWI) with Distributed Acoustic Sensing (DAS) on a subglacial volcano: Grímsvötn, Iceland</b><br>Sara Klaasen, Sebastian Noe, Solvi Thrastarson, Yesim Cubuk-Sabuncu, Kristin Jonsdottir, Jonas Igel, Andreas Fichtner<br>(AGU Annual Meeting)                                                  | 2023 |

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| <b>Volcano monitoring and modelling using Distributed Acoustic Sensing (DAS) on a sub-glacial volcano: Grímsvötn, Iceland</b>                      | 2023 |
| Sara Klaasen, Solvi Thrastarson, Yesim Cubuk-Sabuncu, Kristin Jonsdottir, Lars Gebraad, Sebastian Noe, Andreas Fichtner<br>(IUGG General Assembly) |      |
| <b>Monitoring glacially covered volcanoes with Fibre-Optic Sensing</b>                                                                             | 2023 |
| Sara Klaasen, Solvi Thrastarson, Yesim Cubuk-Sabuncu, Kristin Jonsdottir, Lars Gebraad, Andreas Fichtner<br>(IAVCEI General Meeting)               |      |
| <b>Volcanic Event Characterization with Fiber-Optic Sensing</b>                                                                                    | 2022 |
| Sara Klaasen, Solvi Thrastarson, Yesim Cubuk-Sabuncu, Kristin Jonsdottir, Lars Gebraad, Andreas Fichtner<br>(AGU Annual Meeting)                   |      |
| <b>Fibre-Optic Sensing for Volcano Monitoring on Grímsvötn, Iceland</b>                                                                            | 2022 |
| Sara Klaasen, Solvi Thrastarson, Yesim Cubuk-Sabuncu, Kristin Jonsdottir, Lars Gebraad, Andreas Fichtner<br>(EGU General Assembly)                 |      |
| <b>Distributed Acoustic Sensing on Volcanoes: Grímsvötn and Fagradalsfjall, Iceland</b>                                                            | 2021 |
| Sara Klaasen, Solvi Thrastarson, Kristin Jonsdottir, Yesim Cubuk-Sabuncu, Patrick Paitz, Andreas Fichtner<br>(AGU Annual Meeting)                  |      |
| <b>Combining Distributed Acoustic Sensing and Beamforming in a Volcanic Environment on Mount Meager, British Columbia</b>                          | 2021 |
| Sara Klaasen, Patrick Paitz, Jan Dettmer, Andreas Fichtner<br>(SSA General Meeting)                                                                |      |
| <b>Combining Distributed Acoustic Sensing and Beamforming in a Volcanic Environment on Mount Meager, British Columbia</b>                          | 2021 |
| Sara Klaasen, Patrick Paitz, Jan Dettmer, Andreas Fichtner<br>(EGU General Assembly)                                                               |      |
| <b>Distributed Acoustic Sensing for the Exploration of the Mount Meager Volcanic Complex, British Columbia, Canada</b>                             | 2020 |
| Sara Klaasen, Andreas Fichtner, Patrick Paitz, Jan Dettmer<br>(AGU Annual Meeting)                                                                 |      |

## Invited Talks

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| <b>Keck observatory workshop</b>                                                              | 2026 |
| SSA optical seismology and the next era of seismic sensing, Kona<br>(Pre-conference workshop) |      |
| <b>Frontiers in Fiber-Optics Sensing in Seismology</b>                                        | 2025 |
| IASPEI-IAGA, Lisbon<br>(Conference session)                                                   |      |
| <b>ECR Keynote talk</b>                                                                       | 2025 |
| IAVCEI, Geneva<br>(Conference keynote)                                                        |      |

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| <b>Seminar</b><br>Stanford University, San Francisco<br>(Seminar)                                                                    | 2025 |
| <b>Seminar</b><br>University of Washington, online<br>(Seminar)                                                                      | 2024 |
| <b>Session: An innovative photonic vision of volcanoes and geothermal systems</b><br>SSA Photonic Seismology, Vancouver<br>(Session) | 2024 |
| <b>Colloquium</b><br>CSIRO, Perth<br>(Seminar)                                                                                       | 2023 |
| <b>Geophysics Weekly Meeting</b><br>Australian National University, Canberra<br>(Seminar)                                            | 2023 |
| <b>Seminar</b><br>University of Canterbury, Christchurch<br>(Seminar)                                                                | 2023 |
| <b>Workshop on Distributed Acoustic Sensing</b><br>AGU, online<br>(Pre-conference workshop)                                          | 2021 |

## Service

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**Convening conference sessions**  
IAVCEI, AGU

**Peer-review**  
Seismica, SRL

## Languages

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**Dutch**  
native

**English**  
C2

**German**  
B2

**French**  
B1